

TOSHIBA MOS MEMORY PRODUCTS

256K BIT (32K WORD × 8 BIT) CMOS MASK ROM
SILICON GATE MOS

TC53257P
TC53257F

DESCRIPTION

The TC53257P/F is a 262,144 bit read only memory organized as 32,768 words by 8 bits with a low bit cost, this being suitable for use in program memory of microprocessor, and in character generator. The TC53257P/F using CMOS technology is most suitable for low power applications where bat-

tery operation is required.

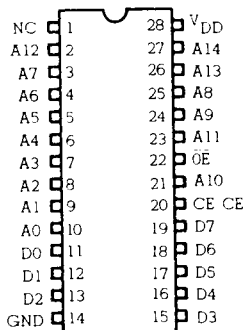
The TC53257P/F has one programmable chip enable input $\overline{CE}/CE$, for device selection and one output enable input ($\overline{OE}$) for fast memory access and output control.

FEATURES

- Single 5V Power Supply
- Access Time: 200ns(Max.)
- Power Dissipation
Operating Current : 25mA(Max.)
Standby Current : 20 μ A(Max.)
- Pin Compatible with 256K EPROM TC57256AD/ADI
- Pin Compatible with 256K OTPROM TC54256AP/AF
- Full Static Operation

- Programmable Chip Enable
- All Inputs and Outputs : TTL Compatible
- Three State Outputs
- Package
Plastic DIP : TC53257P
Plastic FP : TC53257F

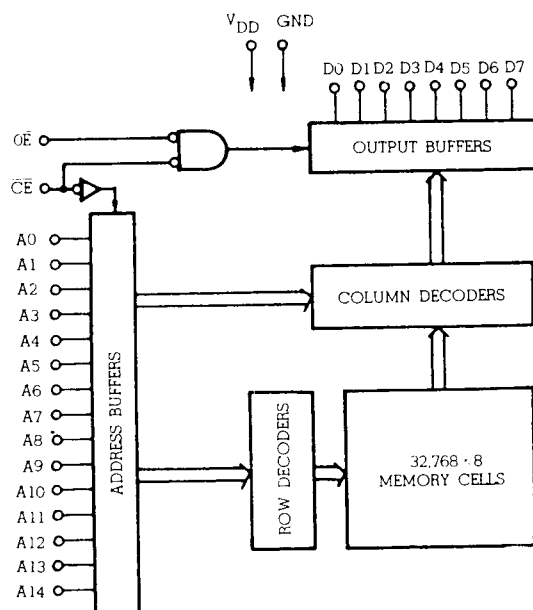
PIN CONNECTION (TOP VIEW)



PIN NAMES

A ₀ ~A ₁₄	Address Inputs
D ₀ ~D ₇	Data Outputs
NC	No connection
$\overline{CE}/CE$	Chip enable input
$\overline{OE}$	Output enable input
V _{DD}	Power supply
GND	Ground

BLOCK DIAGRAM



TC53257P

TC53257F

MAXIMUM RATINGS

SYMBOL	ITEM	RATING	UNITS
V_{DD}	Power Supply Voltage	$-0.5 \sim 7.0$	V
V_{IN}	Input Voltage	$-0.5 \sim 7.0$	V
V_{OUT}	Output Voltage	$0 \sim V_{DD}$	V
P_D	Power Dissipation	$1.0 \ 0.6^*$	W
T_{STG}	Storage Temperature	$-55 \sim 150$	°C
T_{OPR}	Operating Temperature	$-40 \sim 85$	°C
T_{SOLDER}	Soldering Temperature·Time	$260 \cdot 10$	°C·sec

Note : *Plastic FP

D. C. OPERATING CONDITINS (Ta = -40~85°C)

SYMBOL	PARAMETER	MIN.	TYP.	MAX.	UNIT
V_{DD}	Power Supply Voltage	4.5	5.0	5.5	V
V_{IH}	Input High Voltage	2.2	—	$V_{DD} + 0.3$	V
V_{IL}	Input Low Voltage	-0.3	—	0.8	V

D. C. and OPERATING CHARACTERISTICS (Ta = -40~85°C, $V_{DD} = 5V \pm 10\%$)

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
I_{IL}	Input Leakage Current	$V_{IN} = 0V \sim V_{DD}$	—	± 1.0	μA
I_{LO}	Output Leakage Current	$\overline{CE} = V_{IH}$ or $\overline{OE} = V_{IH}$ $V_{OUT} = 0V \sim V_{DD}$	—	± 5.0	μA
I_{OH}	Output High Current	$V_{OH} = 2.4V$	-1.0	—	mA
I_{OL}	Output Low Current	$V_{OL} = 0.4V$	3.2	—	mA
I_{DDS1}	Standby Current	$\overline{CE} = V_{IH}$ $\overline{CE} = V_{IL}$	—	2	mA
I_{DDS2}	Standby Current	$\overline{CE} = V_{DD} - 0.2V$, $\overline{CE} = 0.2V$	—	20	μA
I_{DDO1}	Operating Current	$V_{IH} = V_{IH \ VIL}$, $t_{CYCLE} = 200ns$	—	40	mA
I_{DDO2}		$V_{IN} = V_{DD} - 0.2V \ 0.2V$, $t_{CYCLE} = 200ns$	—	25	mA

CAPACITANCE

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
C_{IN}	Input Capacitance	$f = 1MHz$, $T_a = 25^\circ C$	—	8	pF
C_{OUT}	Output Capacitance	$f = 1MHz$, $T_a = 25^\circ C$	—	10	pF

Note : This parameter is periodically sampled and is not 100% tested.

A. C. CHARACTERISTICS

($T_a = -40 \sim 85^\circ\text{C}$, $V_{DD} = 5V \pm 10\%$)

SYMBOL	PARAMETER	MIN.	MAX.	UNIT
t_{CYC}	Cycle Time	200	—	ns
t_{ACC}	Access Time	—	200	ns
t_{CE}	Chip Enable Access Time from $\overline{CE}/CE$	—	200	ns
t_{OE}	Output Enable Access Time from $\overline{OE}$	—	70	ns
t_{CED}, t_{OED}	Output Disable Time from $\overline{CE}/\overline{CE}$, $\overline{OE}$	0	60	ns
t_{OH}	Output Hold Time	0	—	ns

A. C. TEST CONDITIONS

Output Load : 100pF + 1TTL

Input Levels : 0.6V, 2.4V

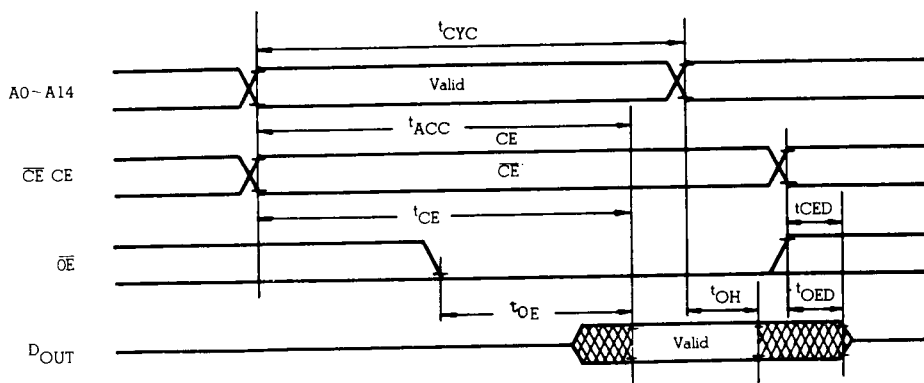
Timing Measurement Reference Levels

Input : 0.8V, 2.2V

Output : 0.8V, 2.2V

Input Rise and Fall Time : 5ns

TIMING WAVEFORMS



OPERATION MODE

H : V_{IH} , L : V_{IL} , * : V_{IH} or V_{IL}

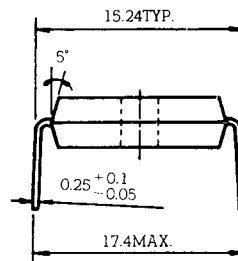
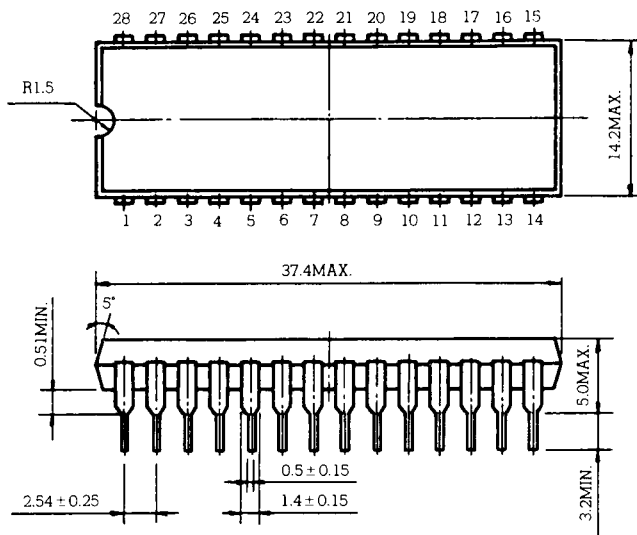
MODE	$\overline{CE}(CE)$	$\overline{OE}$	$A_0 \sim A_{14}$	Outputs	Power
Read	L(H)	L	Valid	Data out	Operating
Output Deselect	L(H)	H	*	High-Z	Operating
	H(L)	*	*		Standby

TC53257P TC53257F

OUTLINE DRAWINGS

● Plastic DIP

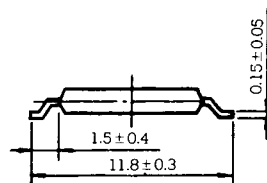
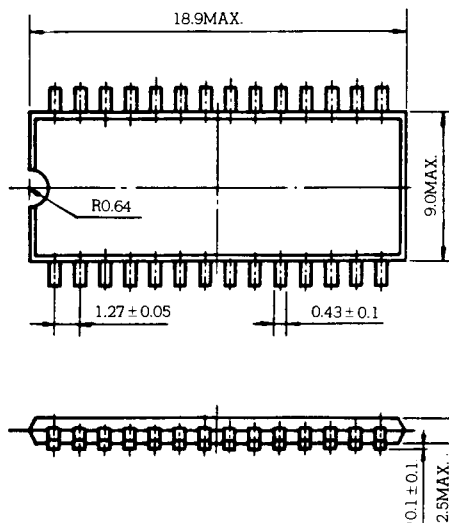
Unit: mm



NOTE : Each lead pitch is 2.54mm.

All leads are located within 0.25mm of their true longitudinal position with respect to No. 1 and No. 28 leads.

● Plastic FP



NOTE : Each lead pitch is 1.27mm.

All leads are located within 0.12mm of their true longitudinal position with respect to No. 1 and No. 28 leads.